INCH-POUND
MIL-DTL-24643/53F
1 October 2009
SUPERSEDING
MIL-DTL-24643/53E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 3000 VOLTS, TYPES LSSRW, LSDRW, AND LSTRW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight

First	-	Copper conductor, coated or uncoated 14 AWG, Class B stranding.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Insulation wall thickness 0.080 inch minimum average. Standard identification code shall be applied by Method 1 or Method 3. If Method 1 is used, the insulation color shall be black.
Fourth	-	The required two or three conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm well-rounded assembly.

NOTE: For LSSRW, omit fourth.

Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket. Minimum average wall thickness of 0.050 inch.

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TABLE I. Details.

Military part no. M24643/53	Cable type	Number of conductors	Overall diameter max. (inch)	Conductor resistance (ohms)
-01UN	LSSRW	1	0.400	2.69
-02UN	LSDRW	2	0.670	2.75
-03UN	LSTRW	3	0.710	2.75

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25°C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor (Types LSDRW and LSTRW only)	8000
Conductor to ground (Type LSSRW only)	8000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor (Types LSDRW and LSTRW only)	9500
Conductor to ground (Type LSSRW only)	9500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300

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Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor), Method 1 only (Types LSDRW and LSTRW only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Cable Sealant removability (Types LSDRW and LSTRW only)	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50

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Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No Failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSSRW	2500
LSDRW and LSTRW	2000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-057)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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